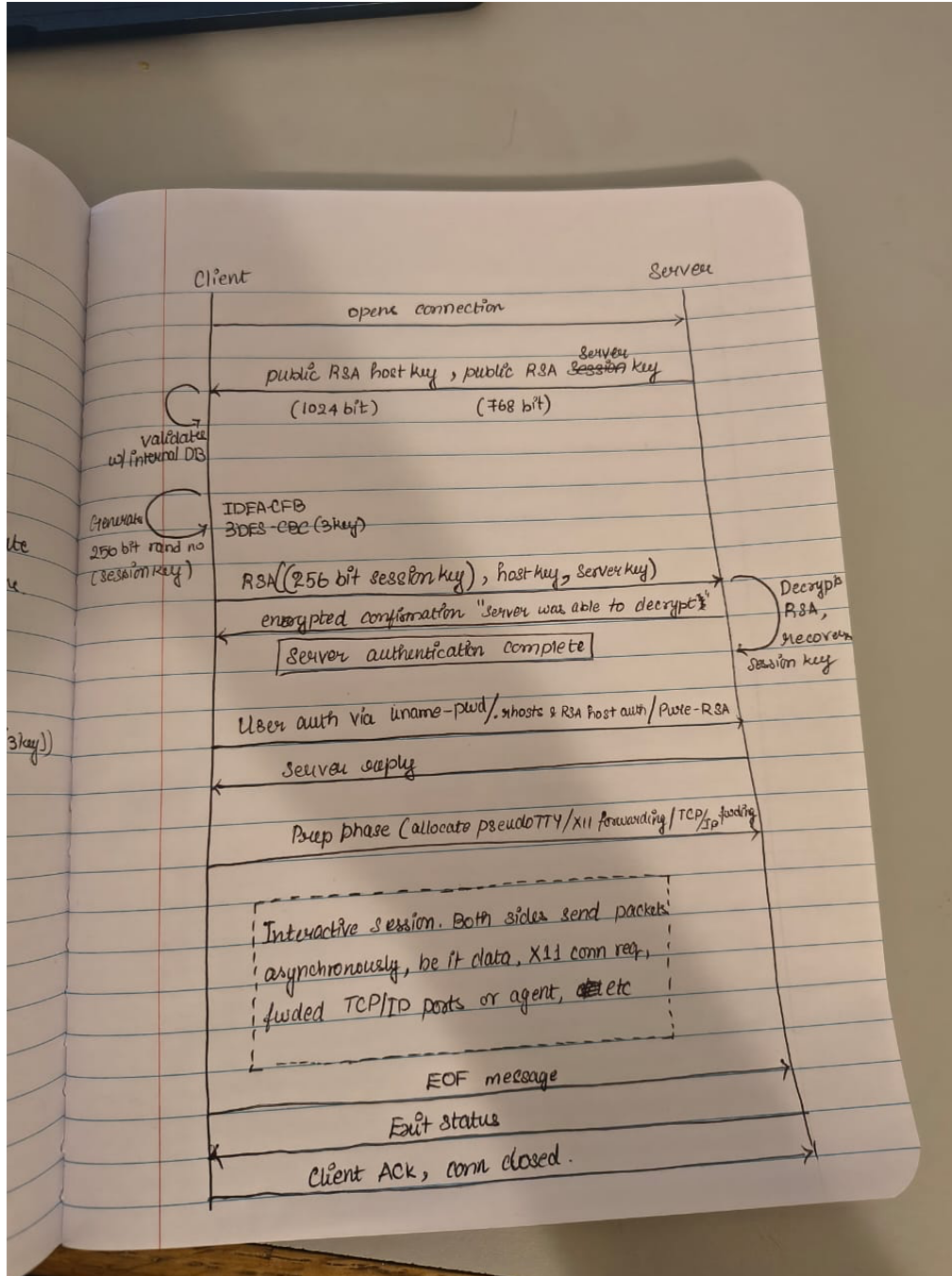


SSH Implementation and MITM Attack

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Part - 01: SSH Overview between Client and Server, from the reference paper



Outputs taken during testing:

Client:

```
Terminal Shell Edit View Window Help
suryapra - skp6318@e5-cse-135-15:~/Coursework/cse543-CompSec/Assignment-1/implementation/01-ssh-implementation-og-client - ssh skp6318@e5-cse-135-15.cse.psu.edu - 208x61
...implementation-og-server - ssh skp6318@e5-cse-135-13.cse.psu.edu ...mentation/02-ssh-mitm - ssh skp6318@e5-cse-135-15.cse.psu.edu ...lementation-og-client - ssh skp6318@e5-cse-135-15.cse.psu.edu +

e5-cse-135-15 165 make clean && make
rm -f *.o *.so cse543-pi cse543-pi-server libcse543-crlib.a
gcc -c -I. -I/usr/include/ -g -Wall cse543-pi.c -o cse543-pi.o
gcc -c -I. -I/usr/include/ -g -Wall cse543-proto.c -o cse543-proto.o
gcc -c -I. -I/usr/include/ -g -Wall cse543-network.c -o cse543-network.o
gcc -c -I. -I/usr/include/ -g -Wall cse543-ssl.c -o cse543-ssl.o
gcc -c -I. -I/usr/include/ -g -Wall cse543-util.c -o cse543-util.o
ar rc libcse543-crlib.a cse543-proto.o cse543-network.o cse543-ssl.o cse543-util.o
ranlib libcse543-crlib.a
gcc -g -L. -L/usr/lib64/ cse543-pi.o -lcrypto -lm -lcse543-crlib -o cse543-pi
gcc -c -I. -I/usr/include/ -g -Wall cse543-pi.c -DCSE543_PROTOCOL_SERVER -o cse543-pi-server.o
gcc -g -L. -L/usr/lib64/ cse543-pi-server.o -lcrypto -lm -lcse543-crlib -o cse543-pi-server
e5-cse-135-15 173 cat textfile.txt
Some Content and Stuff
Updated Content and Stuff

halooooo
Yoyo

Yoyoyoyoe5-cse-135cse543-pi textfile.txt 130.283.16.32
Transfer beginning, file [textfile.txt]
Client connected to address [130.283.16.32/9168], successful ...

file name: textfile.txt
Block is: 73 bytes ...
0000 - 53 6f 64 65 28 43 6f 6e-7a 65 6e 7a 20 61 6e 64 Some Content and
0010 - 20 53 74 75 66 66 0a 55-70 64 61 74 65 64 20 43 Stuff,Updated C
0020 - 6f 64 74 65 6e 7a 20 61-6e 64 20 53 74 75 66 6e Content and Stuff
0030 - 0a 0a 68 65 6c 6f 6f 6f-6f 6a 59 6f 79 6f 0a ..halooooo.Yoyo.
0040 - 0a 59 6f 79 6f 79 6f 79-6f
e5-cse-135-15 193 bytes ...
e5-cse-135-15 193
e5-cse-135-15 193
```

Server:

```
Terminal Shell Edit View Window Help
suryapra - skp6318@e5-cse-135-13:~/Coursework/cse543-CompSec/Assignment-1/implementation/01-ssh-implementation-og-server - ssh skp6318@e5-cse-135-13.cse.psu.edu - 208x61
...implementation-og-server - ssh skp6318@e5-cse-135-13.cse.psu.edu ...mentation/02-ssh-mitm - ssh skp6318@e5-cse-135-15.cse.psu.edu ...lementation-og-client - ssh skp6318@e5-cse-135-15.cse.psu.edu +

e5-cse-135-13 278 make clean && make
rm -f *.o *.so cse543-pi cse543-pi-server libcse543-crlib.a
gcc -c -I. -I/usr/include/ -g -Wall cse543-pi.c -o cse543-pi.o
gcc -c -I. -I/usr/include/ -g -Wall cse543-proto.c -o cse543-proto.o
gcc -c -I. -I/usr/include/ -g -Wall cse543-network.c -o cse543-network.o
gcc -c -I. -I/usr/include/ -g -Wall cse543-ssl.c -o cse543-ssl.o
gcc -c -I. -I/usr/include/ -g -Wall cse543-util.c -o cse543-util.o
ar rc libcse543-crlib.a cse543-proto.o cse543-network.o cse543-ssl.o cse543-util.o
ranlib libcse543-crlib.a
gcc -g -L. -L/usr/lib64/ cse543-pi.o -lcrypto -lm -lcse543-crlib -o cse543-pi
gcc -c -I. -I/usr/include/ -g -Wall cse543-pi.c -DCSE543_PROTOCOL_SERVER -o cse543-pi-server.o
gcc -g -L. -L/usr/lib64/ cse543-pi-server.o -lcrypto -lm -lcse543-crlib -o cse543-pi-server
e5-cse-135-13 285 rm shared/*
e5-cse-135-13 293 ls shared/
e5-cse-135-13 303 cse543-pi-server mykey.pem raspub.pem
Server binding to port [9168], successful ...
*** Test RSA encrypt and decrypt. ***
Ciphertext is:
0000 - aa e9 8a 21 cc ea 5b 67-ba ac fa 0c 82 c9 38 7b ..Tl..lg...R.8f
0010 - 03 0d b4 4e ec fa 58 94-56 72 a9 a9 46 46 b4 99 ...N..P.Vr..FF..
Msg: help me, Mr. Wizard!

*** Test AES encrypt and decrypt. ***
Ciphertext is:
0000 - 16 ef fc 66 58 df a8 bf-8b 24 aa 79 0a 2a 0e 42 ...fX....$.y.*.B
0010 - b2 9b 3e 43 ..3C
Tag is:
0000 - 7a 63 ea 9a b2 d8 34 4b-7a ca 5f 9e 28 ac 12 c4 VC....AK2...{...
Msg: Help me, Mr. Wizard!
fname: ./shared/textfile.txtReceiving file [./shared/textfile.txt] ..
Decrypted block is:
0000 - 53 6f 64 65 28 43 6f 6e-7a 65 6e 7a 20 61 6e 64 Some Content and
0010 - 20 53 74 75 66 66 0a 55-70 64 61 74 65 64 20 43 Stuff,Updated C
0020 - 6f 64 74 65 6e 7a 20 61-6e 64 20 53 74 75 66 6e Content and Stuff
0030 - 0a 0a 68 65 6c 6f 6f 6f-6f 6a 59 6f 79 6f 0a ..halooooo.Yoyo.
0040 - 0a 59 6f 79 6f 79 6f 79-6f
Received/written 73 bytes ...
Total bytes [73].
^C
e5-cse-135-13 315
e5-cse-135-13 315 cat shared/textfile.txt
Some Content and Stuff
Updated Content and Stuff

halooooo
Yoyo

Yoyoyoyoe5-cse-135-13 325
```

Part 2: MITM Attack

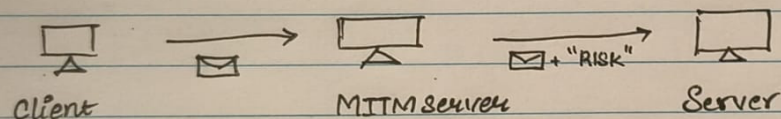
Clarification on Approach:

MITM attack.

The client and the MITM server are present on server-15.

The intended server is run as a standalone instance, on ~~machine 3~~ server-13.

The flow is as follows:



- i) The MITM server receives the file from client.
- ii) The socket is closed (ensures full file is present)
- iii) MITM server then uses client-secure-transfer to act as client and send the file to the server
↳ modified

Output Taken during Testing

Client:

```
Terminal Shell Edit View Window Help
suryapra ~ skp6318@e5-cse-135-15:~/Coursework/cse543-CompSec/Assignment-1/Implementation/01-ssh-implementation-og-client - ssh skp6318@e5-cse-135-15.cse.psu.edu - 208x61
..Implementation-og-server - ssh skp6318@e5-cse-135-13.cse.psu.edu ..mentation/02-ssh-mitm - ssh skp6318@e5-cse-135-15.cse.psu.edu ..mentation-og-client - ssh skp6318@e5-cse-135-15.cse.psu.edu

e5-cse-135-15 125 make clean && make
rm -f *.o *.a cse543-p1 cse543-p1-server libcse543-crlib.a
gcc -c -I. -I/usr/include/ -g -Wall cse543-p1.c -o cse543-p1.o
gcc -c -I. -I/usr/include/ -g -Wall cse543-protoc.c -o cse543-protoc.o
gcc -c -I. -I/usr/include/ -g -Wall cse543-network.c -o cse543-network.o
gcc -c -I. -I/usr/include/ -g -Wall cse543-sal.c -o cse543-sal.o
gcc -c -I. -I/usr/include/ -g -Wall cse543-util.c -o cse543-util.o
ar rc libcse543-crlib.a cse543-protoc.o cse543-network.o cse543-sal.o cse543-util.o
ranlib libcse543-crlib.a
gcc -g -L. -L/usr/lib64/ cse543-p1.o -lcrypto -lm -lcse543-crlib -o cse543-p1
gcc -c -I. -I/usr/include/ -g -Wall cse543-p1.c -DCSE543_PROTOCOL_SERVER -o cse543-p1-server.o
gcc -g -L. -L/usr/lib64/ cse543-p1-server.o -lcrypto -lm -lcse543-crlib -o cse543-p1-server
e5-cse-135-15 138 cse543-p1 mitm-textfile.txt 138.283.16.34
Transfer beginning, file [mitm-textfile.txt]
Client connected to address [138.283.16.34/9165], successful ...

file name: mitm-textfile.txt
Block is: 56 bytes ...
0000 - 48 69 20 48 61 60 6f 20-57 61 60 63 6f 6d 0a 0a Hi Halo Welcom..
0010 - 54 68 69 73 20 69 73 20-46 69 79 61 72 0a 0a 4e This is Fiyar..N
0020 - 65 77 20 4c 69 6e 65 0a-0a 4c 61 7a 79 20 42 72 ew Line..Lazy Br
0030 - 6f 77 6e 28 45 6f 78 20- own Fox
e5-cse-135-15 148 cat mitm-textfile.txt
Hi Halo Welcom
This is Fiyar
New Line
Lazy Brown Fox e5-cse-135-15 155
```

Original Server:

```
Terminal Shell Edit View Window Help
suryapra ~ skp6318@e5-cse-135-15:~/Coursework/cse543-CompSec/Assignment-1/Implementation/02-ssh-mitm - ssh skp6318@e5-cse-135-15.cse.psu.edu - 208x61
..Implementation-og-server - ssh skp6318@e5-cse-135-13.cse.psu.edu ..mentation/02-ssh-mitm - ssh skp6318@e5-cse-135-15.cse.psu.edu ..mentation-og-client - ssh skp6318@e5-cse-135-15.cse.psu.edu

e5-cse-135-15 164 make clean && make
rm -f *.o *.a cse543-p2 cse543-p2-server libcse543-crlib.a
gcc -c -I. -I/usr/include/ -g -Wall cse543-p2.c -o cse543-p2.o
gcc -c -I. -I/usr/include/ -g -Wall cse543-protoc.c -o cse543-protoc.o
gcc -c -I. -I/usr/include/ -g -Wall cse543-network.c -o cse543-network.o
gcc -c -I. -I/usr/include/ -g -Wall cse543-sal.c -o cse543-sal.o
gcc -c -I. -I/usr/include/ -g -Wall cse543-util.c -o cse543-util.o
ar rc libcse543-crlib.a cse543-protoc.o cse543-network.o cse543-sal.o cse543-util.o
ranlib libcse543-crlib.a
gcc -g -L. -L/usr/lib64/ cse543-p2.o -lcrypto -lm -lcse543-crlib -o cse543-p2
gcc -c -I. -I/usr/include/ -g -Wall cse543-p2.c -DCSE543_PROTOCOL_SERVER -o cse543-p2-server.o
gcc -g -L. -L/usr/lib64/ cse543-p2-server.o -lcrypto -lm -lcse543-crlib -o cse543-p2-server
e5-cse-135-15 178 cse543-p2-server nyker.pem raspub.pem 138.283.16.32 1
Server binding to port [9165], successful ...

server waiting for FILE_XFER_INIT
fname: ./shared/mitm-textfile.txtReceiving file [./shared/mitm-textfile.txt] ..
Decrypted Block is:
0000 - 48 69 20 48 61 60 6f 20-57 61 60 63 6f 6d 0a 0a Hi Halo Welcom..
0010 - 54 68 69 73 20 69 73 20-46 69 79 61 72 0a 0a 4e This is Fiyar..N
0020 - 65 77 20 4c 69 6e 65 0a-0a 4c 61 7a 79 20 42 72 ew Line..Lazy Br
0030 - 6f 77 6e 28 45 6f 78 20- own Fox
Received/written 56 bytes ...
Total bytes [56].
Client connected to address [138.283.16.32/9165], successful ...

file name: ./shared/mitm-textfile.txt
Block is: 85 bytes ...
0000 - 48 69 20 48 61 60 6f 20-57 61 60 63 6f 6d 0a 0a Hi Halo Welcom..
0010 - 54 68 69 73 20 69 73 20-46 69 79 61 72 0a 0a 4e This is Fiyar..N
0020 - 65 77 20 4c 69 6e 65 0a-0a 4c 61 7a 79 20 42 72 ew Line..Lazy Br
0030 - 6f 77 6e 28 45 6f 78 20- own Fox
0040 - 41 4e 53 46 45 52 20 49-53 28 55 4e 44 45 52 28 ANSWER IS UNDER
0050 - 52 49 53 4b 0a RISK.
e5-cse-135-15 188 cat shared/mitm-textfile.txt
Hi Halo Welcom
This is Fiyar
New Line
Lazy Brown Fox
YOUR TRANSFER IS UNDER RISK
e5-cse-135-15 196
```

MITM Server :

```
Terminal Shell Edit View Window Help
suryapra ~ skp6318@e5-cse-135-13:~/Coursework/cse543-CompSec/Assignment-1/Implementation/01-ssh-implementation-og-server — ssh skp6318@e5-cse-135-13.cse.psu.edu — 208x61
Implementation-og-server — ssh skp6318@e5-cse-135-13.cse.psu.edu ...mentation/02-ssh-mitm — ssh skp6318@e5-cse-135-15.cse.psu.edu ...mentation-og-client — ssh skp6318@e5-cse-135-15.cse.psu.edu +

e5-cse-135-13 238 make clean && make
rm -f *.o *.so cse543-p1 cse543-p1-server libcse543-crlib.a
gcc -c -I. -I/usr/include/ -g -Wall cse543-p1.c -o cse543-p1.o
gcc -c -I. -I/usr/include/ -g -Wall cse543-proto.c -o cse543-proto.o
gcc -c -I. -I/usr/include/ -g -Wall cse543-network.c -o cse543-network.o
gcc -c -I. -I/usr/include/ -g -Wall cse543-sll.c -o cse543-sll.o
gcc -c -I. -I/usr/include/ -g -Wall cse543-util.c -o cse543-util.o
ar rc libcse543-crlib.a cse543-proto.o cse543-network.o cse543-sll.o cse543-util.o
ranlib libcse543-crlib.a
gcc -g -I. -I/usr/lib64/ cse543-p1.o -lcrypto -lm -lcse543-crlib -o cse543-p1
gcc -c -I. -I/usr/include/ -g -Wall cse543-p1.c -DCSE543_PROTOCOL_SERVER -o cse543-p1-server.o
gcc -g -I. -I/usr/lib64/ cse543-p1-server.o -lcrypto -lm -lcse543-crlib -o cse543-p1-server
e5-cse-135-13 245 cse543-p1-server mykey.pem rsapub.pem
Server binding to port [9165], successful ...
*** Test RSA encrypt and decrypt. ***
Ciphertext is:
0000 - 64 9a 44 03 9b 13 f8 66-8a 5e 2b 26 8a b6 95 1c d.D...f.*&...
0010 - 72 9d 17 28 48 aa f3 9c-a3 58 2d d5 f8 cb 53 77 ...lm...X...B.
Mag: help me, Mr. Wizard!

*** Test AES encrypt and decrypt. ***
Ciphertext is:
0000 - 8b 7c 66 ac de ef 17 93-93 63 66 b9 3a bb e1 46 .lf.....cf.i..F
0010 - 54 09 28 f5 T.().
Tag is:
0000 - 58 8a 86 7f 5c d8 6f 62-4c b8 8f 24 d7 76 7b 8a X...obl...$.v(.
Mag: Help me, Mr. Wizard!
fname: ./shared/mitm-textfile.txtReceiving file [./shared/mitm-textfile.txt] ...
Decrypted block is:
0000 - 48 69 28 48 61 6c ef 28-57 61 6c 63 6f 6d 0a 0a Hi Halo Walcom..
0010 - 64 68 69 73 28 69 73 28-46 69 79 61 72 8a 8a 4e This is Piyar..M
0020 - 65 77 28 4c 69 6a 65 8a-8a 4c 63 7a 79 28 45 72 me time..Lazy Br
0030 - 6f 77 6a 28 46 6f 78 28-0a 59 4f 55 52 20 64 52 own Fox .YOUR TR
0040 - 41 4e 53 46 45 52 28 49-53 20 55 4e 44 45 52 20 ANSWER IS UNDER
0050 - 52 49 53 4b 0a RISK.
Received/written 85 bytes ...
Total bytes [85].
^C
e5-cse-135-13 255 cat shared/mitm-textfile.txt
Hi Halo Walcom

This is Piyar

New Line

Lazy Brown Fox
YOUR TRANSFER IS UNDER RISK
e5-cse-135-13 268 █
```

Github Repository:

<https://github.com/Surya-Prasad/CSE-543-Computer-Security/tree/master/Project-01>